

Technology Stack

Phase: Requirement Analysis

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. MERN Stack Overview

Layer	Role	Version / Hosting
M — MongoDB	Database (NoSQL, document-based)	Atlas Cloud Cluster
E — Express.js	Backend Web Framework (REST API)	v4.x with ES Modules
R — React.js	Frontend UI Library (SPA)	v19.x via Vite
N — Node.js	JavaScript Runtime (server-side)	v18.x+

2. Frontend Technologies

Technology	Version	Purpose
React	19.x	Component-based UI library
Vite	Latest	Build tool with fast Hot Module Replacement (HMR)
React Router DOM	v6	Client-side routing (nested + protected routes)
Redux Toolkit	Latest	Global state: cart, auth, products, wishlist slices
React Redux	Latest	React bindings for Redux store
Axios	Latest	HTTP client with JWT Bearer interceptor
Vanilla CSS	—	Custom dark-theme design system with CSS custom properties
Google Fonts	—	Typography: Inter / Poppins

3. Backend Technologies

Technology	Version	Purpose
Node.js	18.x+	JavaScript runtime environment
Express.js	4.x	REST API web framework
ES Modules	—	import/export syntax ("type": "module" in package.json)
jsonwebtoken	Latest	JWT generation and verification
bcryptjs	Latest	Password hashing (salt rounds: 10)
Multer	Latest	Multi-file upload middleware for product images
dotenv	Latest	Environment variable management (.env file)
cors	Latest	Cross-Origin Resource Sharing policy

4. Database

Technology	Version	Purpose
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Latest

Cloud-hosted NoSQL database cluster with auto-backups

Mongoose

7.x

ODM — schema definition, validation, relationships

5. Development Tools

Tool	Purpose
VS Code	Primary code editor with ESLint + Prettier
Git + GitHub	Version control and remote repository (main branch)
Postman	API testing and debugging
MongoDB Compass	Local MongoDB database GUI
npm	Package manager for both client and server
OBS Studio / Loom	Screen recording for demo videos

6. npm Start Scripts

npm run dev — Start both client (port 5173) and server (port 5000) concurrently

npm run dev:server — Start only the backend Express server

npm run dev:client — Start only the Vite frontend dev server

node server/seed.js — Seed the database with sample products and admin user